Klinikum rechts der Isar
Technische Universität München

Multi-Modal Deep Learning

Lecture Series | Summer 2025

Institute for AI and Informatics in Medicine

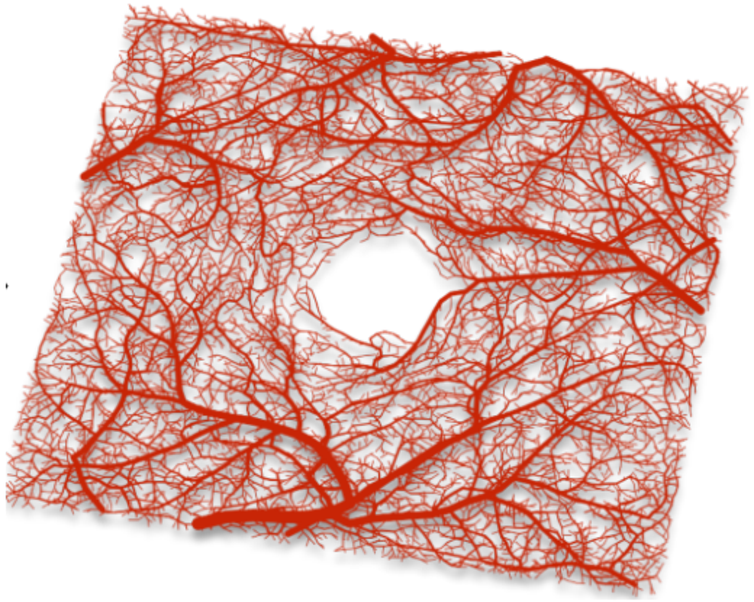
03.06.2025

GRAPHS

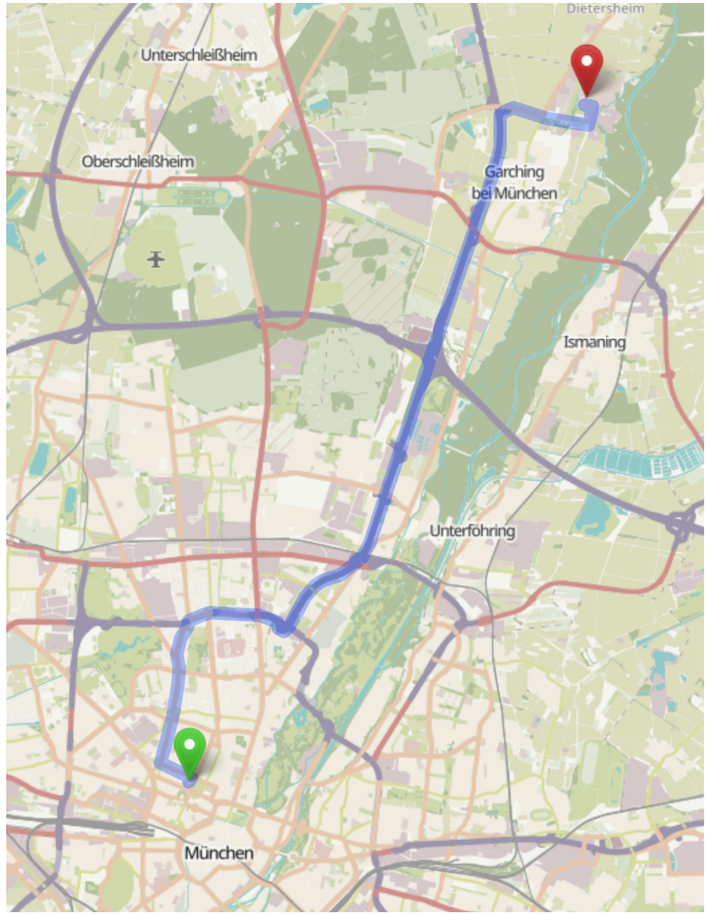
Definition from Wikipedia

"*A graph is a structure consisting of* ***a set of objects where some pairs of the objects are in some sense 'related'***.
The objects are represented by abstractions called vertices and each of the related pairs of vertices is called an edge."

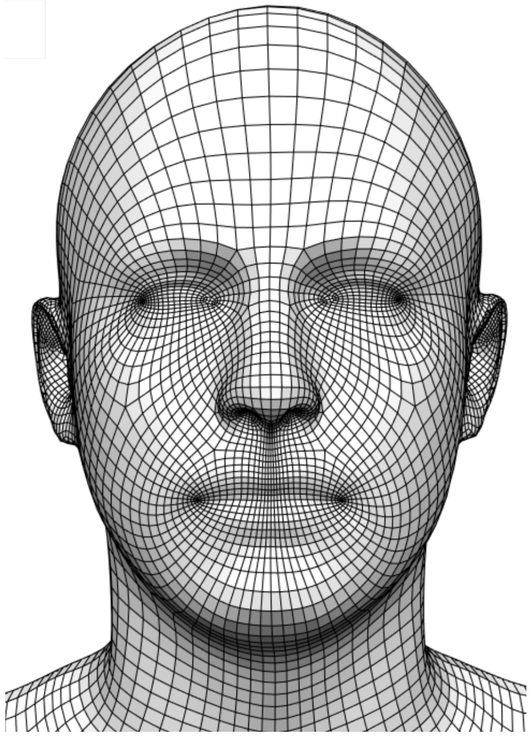
DATA AS GRAPHS



Blood vessel graph

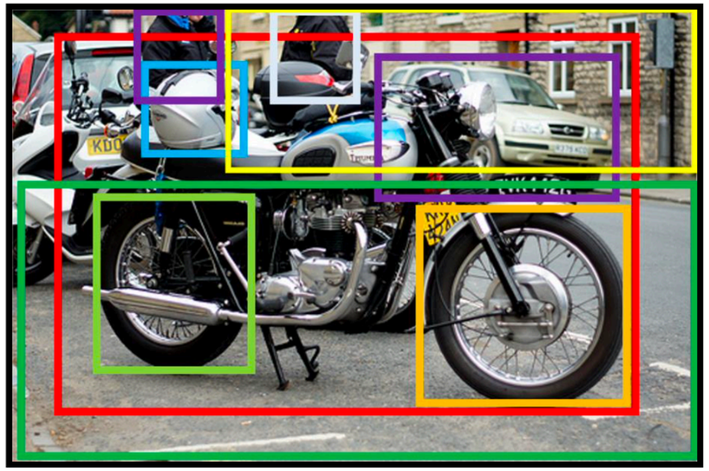


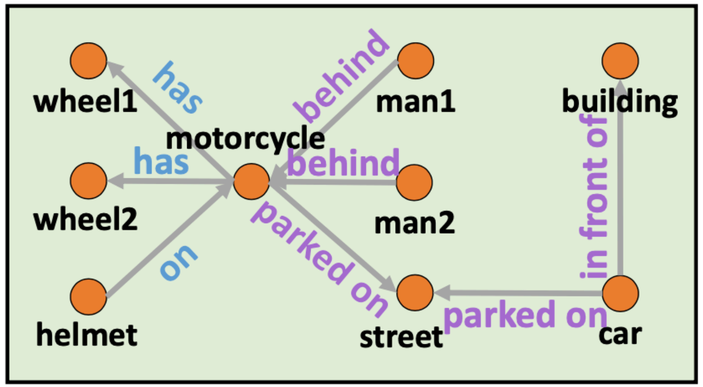
Road network



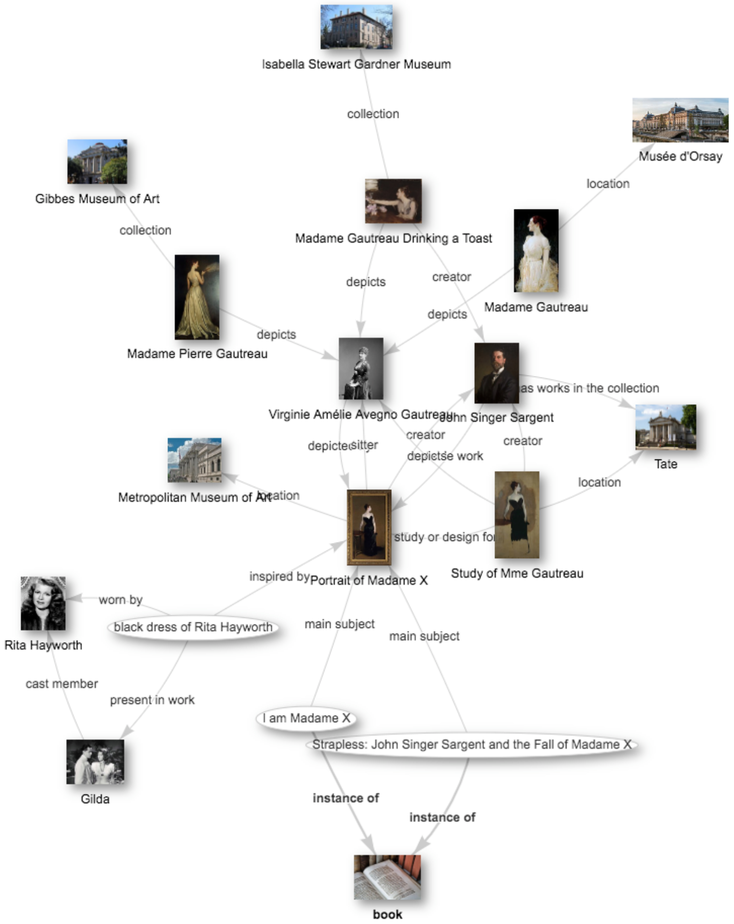
Quadrilateral facial mesh

DATA AS GRAPHS





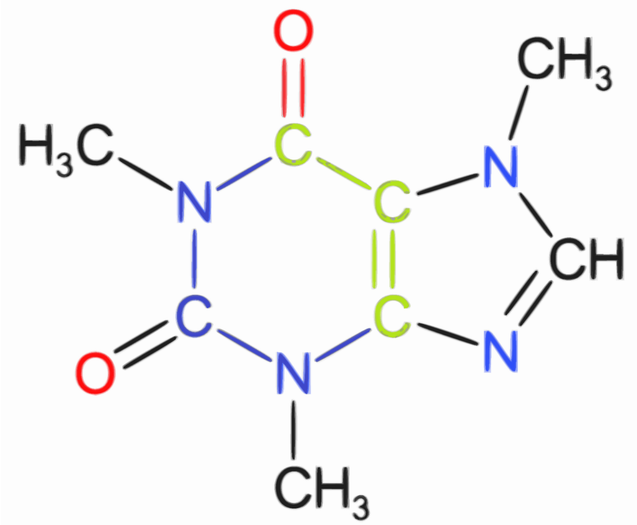
Scene graph



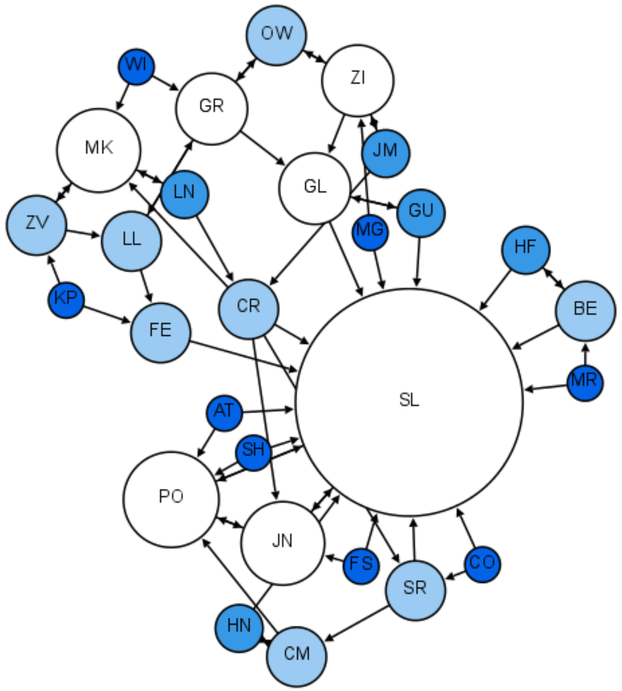
Knowledge graph

Left image taken from Tang, et al. *CVPR* 2020: 3716-3725

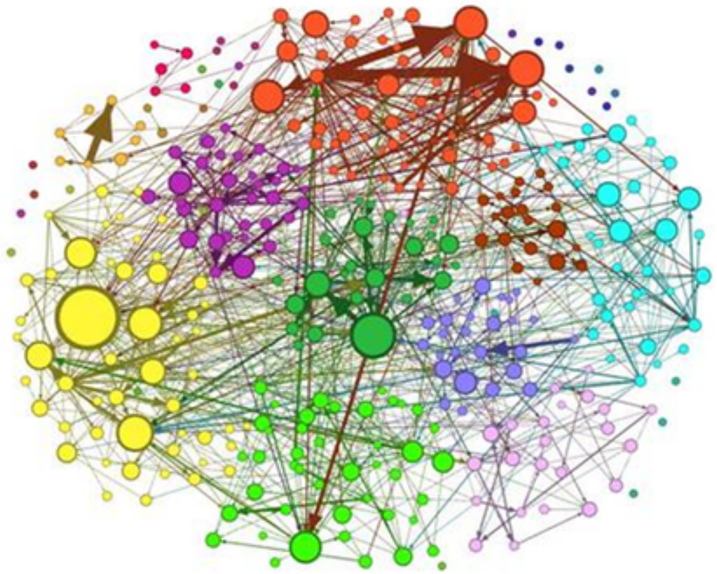
DATA AS GRAPHS



Molecules



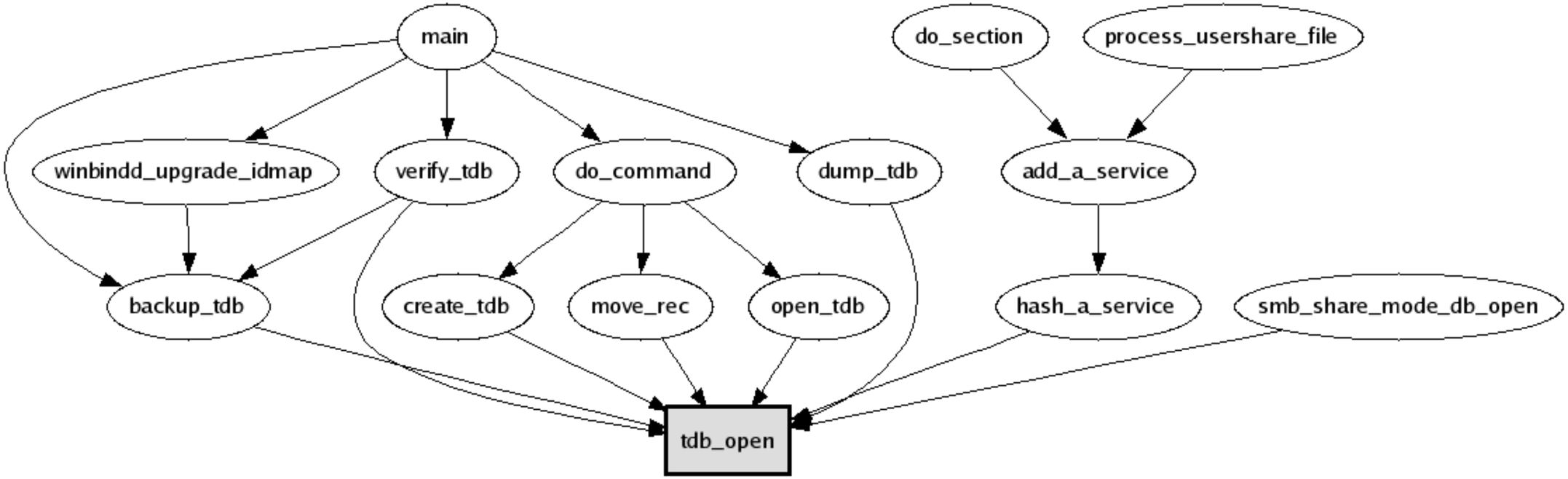
Social graphs



Communication graphs

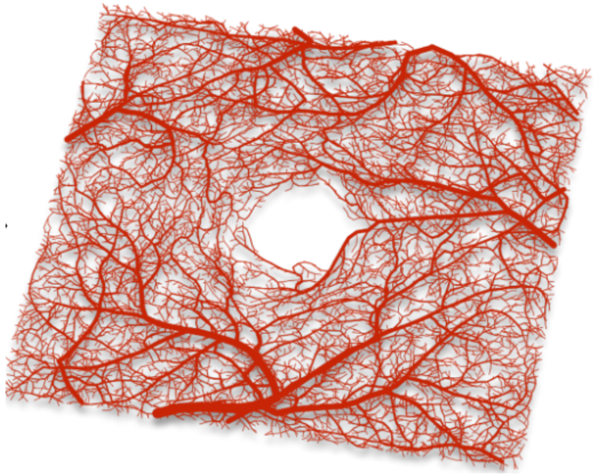
Right image taken from Casnici et al. *PLoS one* 10.8 (2015): e0133712.

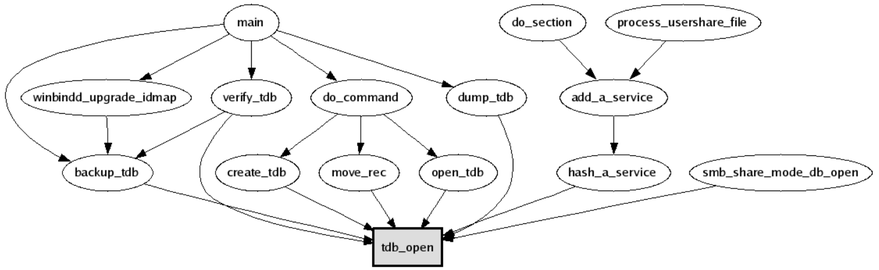
DATA AS GRAPHS

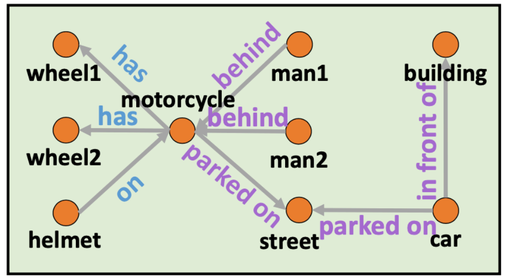


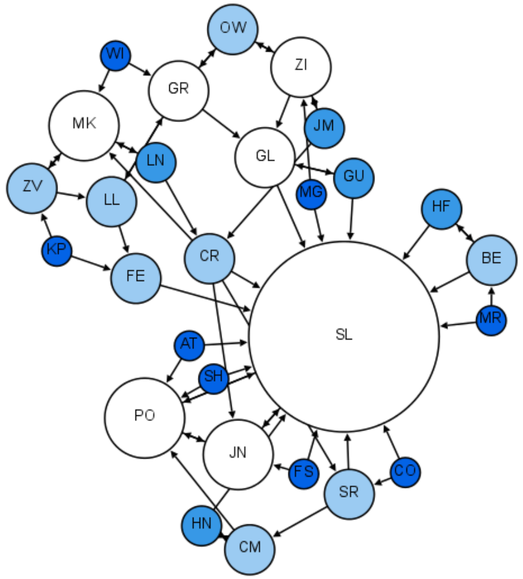
Software code graph

STRUCTURAL VS. RELATIONAL GRAPHS







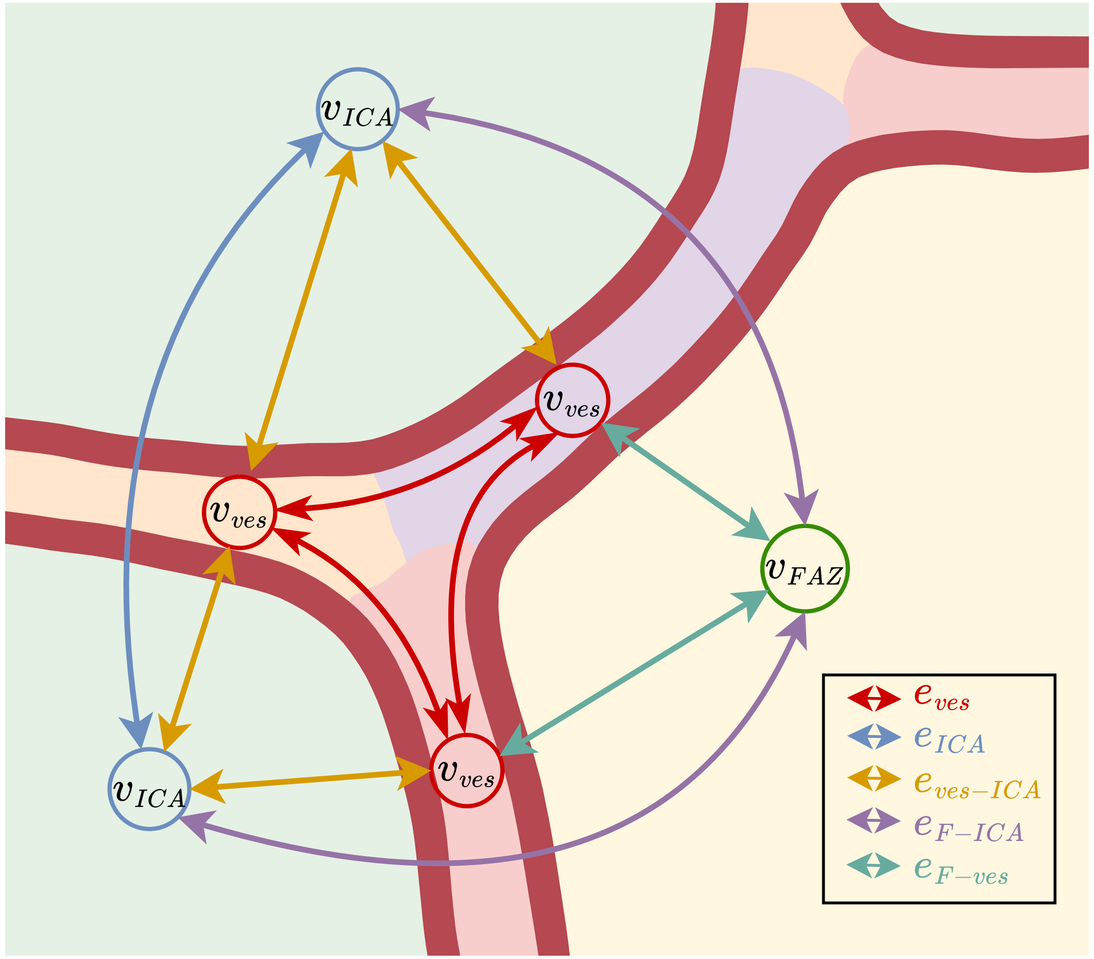


Structural **Relational**

DATA AS GRAPHS

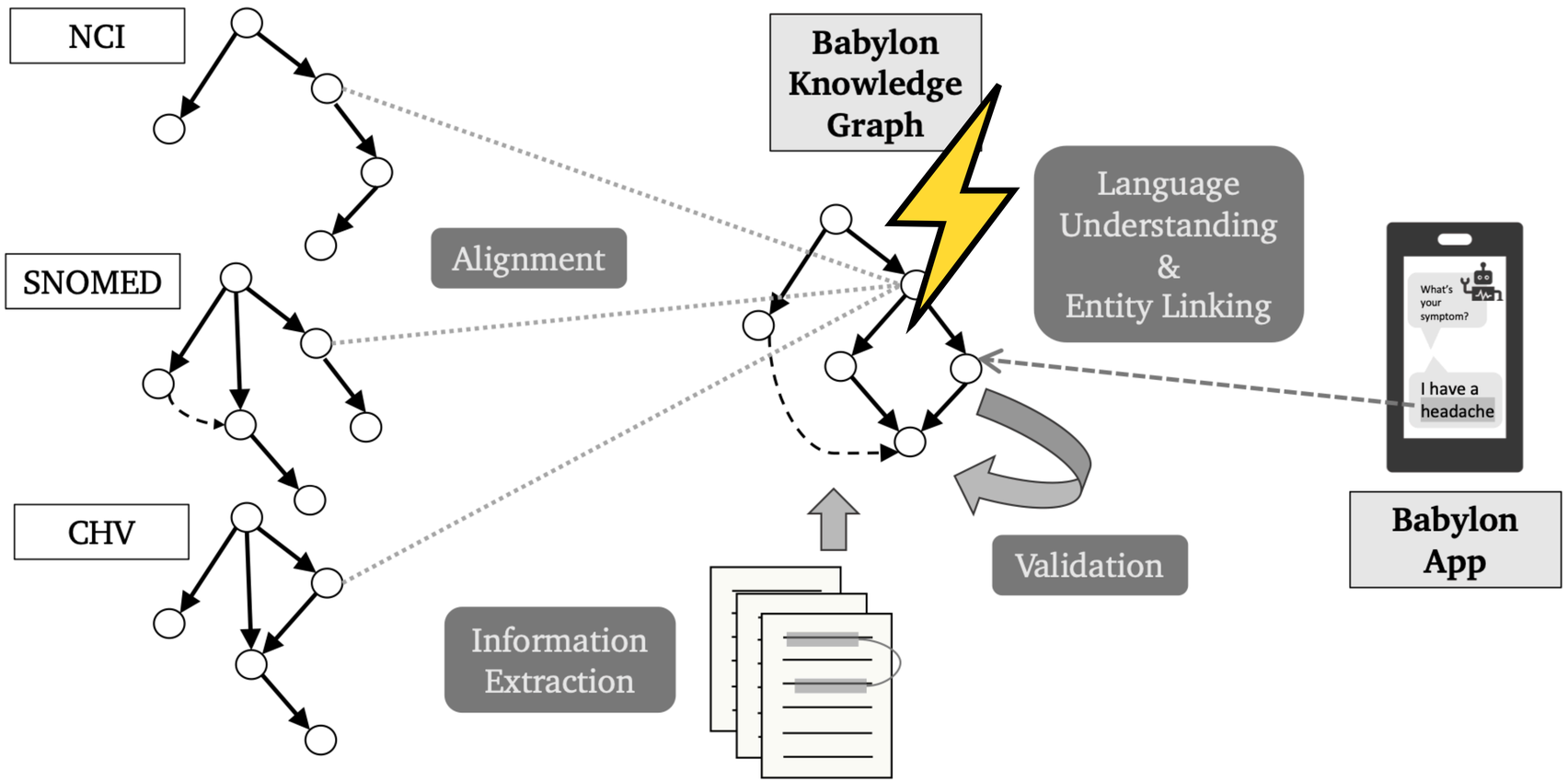
- Many data types are **naturally represented** as graphs
- Examples span **structural, relational and hybrid graphs**
  - Structural graphs focus on the overall graph layout and topology
  - Relational Graphs focus on the relationship between individual entities

DATA AS GRAPHS



Heterogeneous vascular graph

DATA AS GRAPHS



Medical knowledge graph for chatbot backend

Image taken from Schulz et al. SWAT4HCLS 2019

DATA AS GRAPHS

- Many data types are **naturally represented** as graphs
- Examples span **structural, relational and hybrid graphs**
  - Structural graphs focus on the overall graph layout and topology
  - Relational Graphs focus on the relationship between individual entities
- If and how to construct the graph often involves **explicit design choices**
  - In some cases, there is a unique, unambiguous representation
  - In other cases, their are multiple plausible representations
- Choice of graph will determine the nature of the questions you can study
- Graph quality is crucial

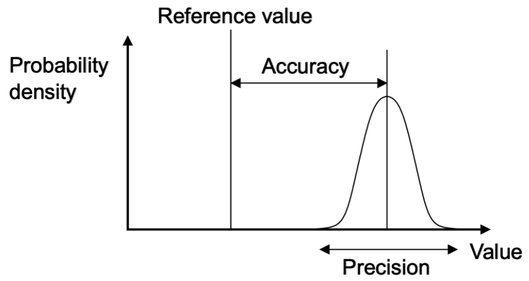
REMINDER: WEEK 1

MEASUREMENT

Measurement is the quantification of attributes of an object or event.

It should be:

- Accurate
- Precise
- Unbiased
- Reproducible



GRAPHS IN MULTIMODAL DEEP LEARNING

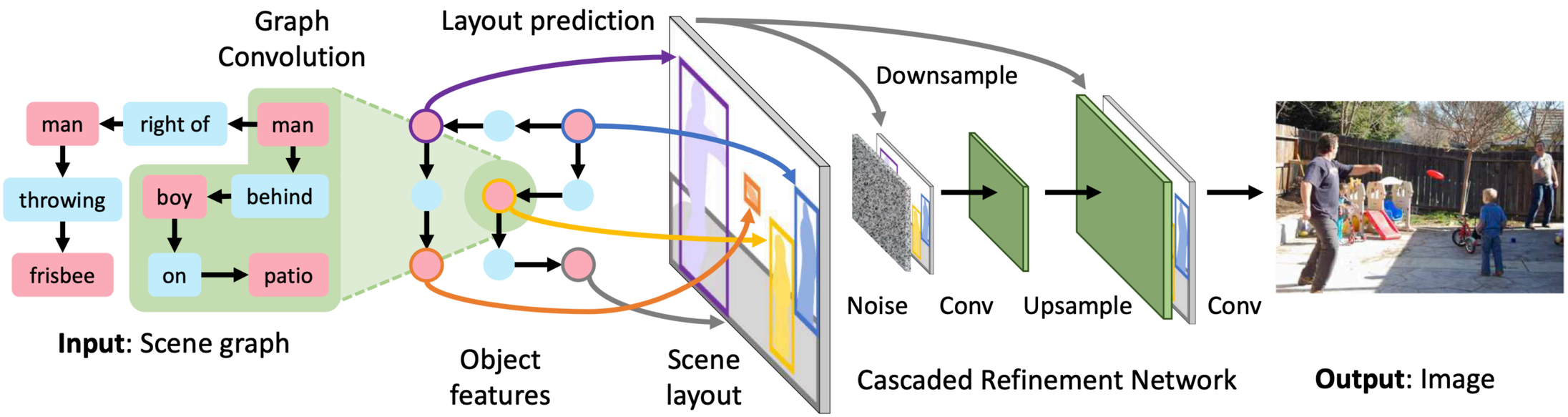
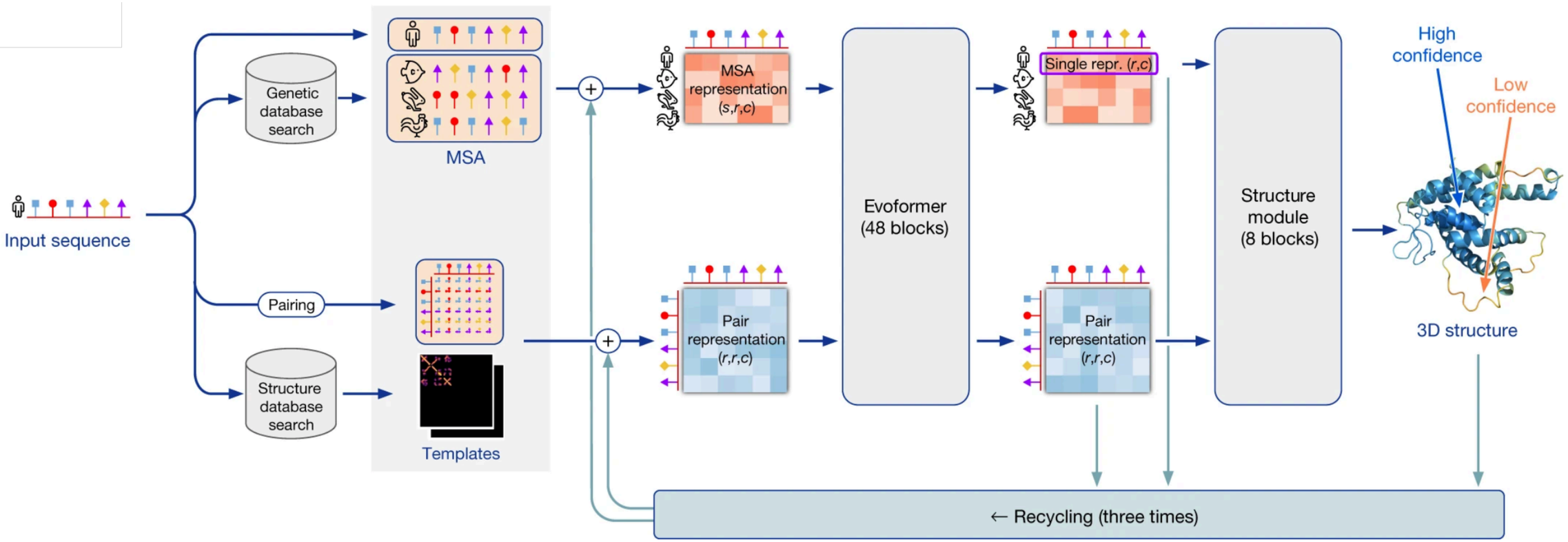


Image generation from scene graphs

Image taken from Johnson et al. *CVPR* 2018: 1219-1228

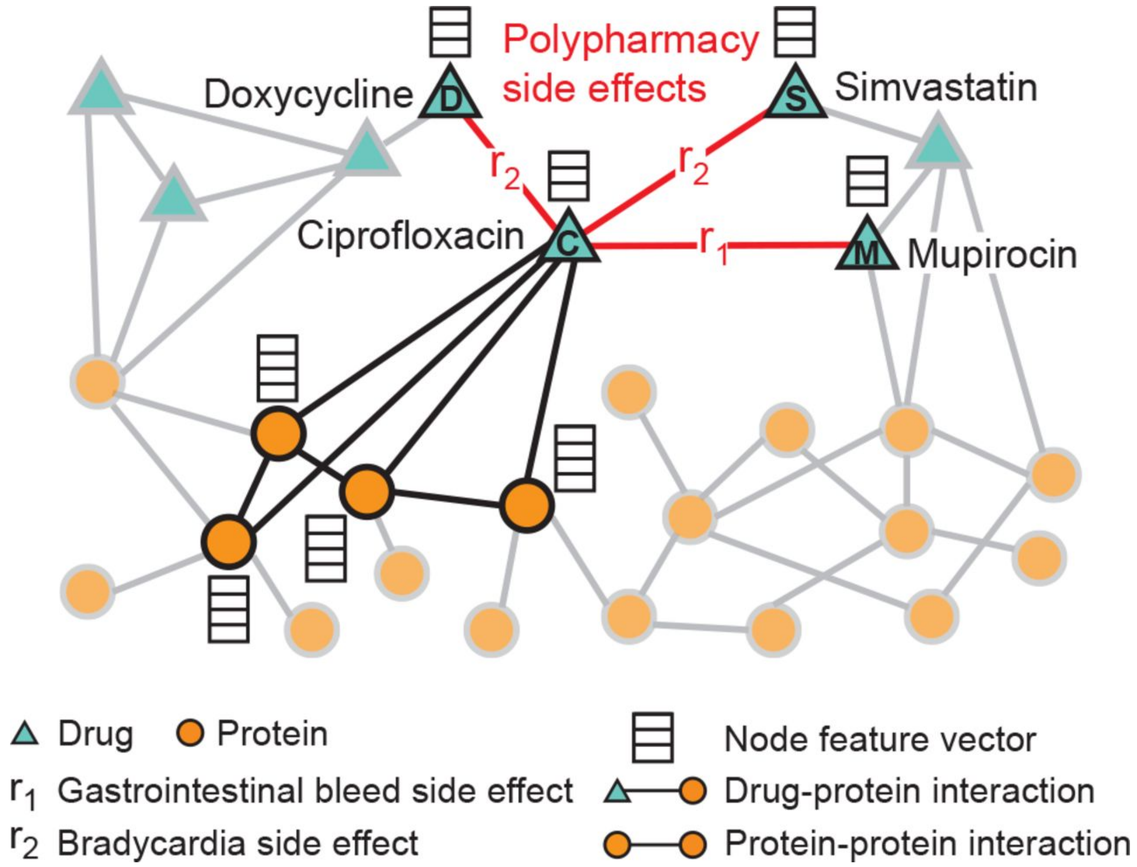
GRAPHS IN MULTIMODAL DEEP LEARNING



Protein sequence to structure

Image taken from Jumper et al. *Nature* 596 (2024): 583–589

GRAPHS IN MULTIMODAL DEEP LEARNING



Drug interaction graph

Image taken from Zitnik et al. *Bioinformatics* 34(13): i457-i466

GRAPHS IN MULTIMODAL DEEP LEARNING

- Graphs are commonly found in **multimodal deep learning**
  - Graph as **input** of a multimodal pipeline
  - Graph as **output** of a multimodal pipeline
  - Graph as **structure to relate** multimodal data embeddings

THIS LECTURE

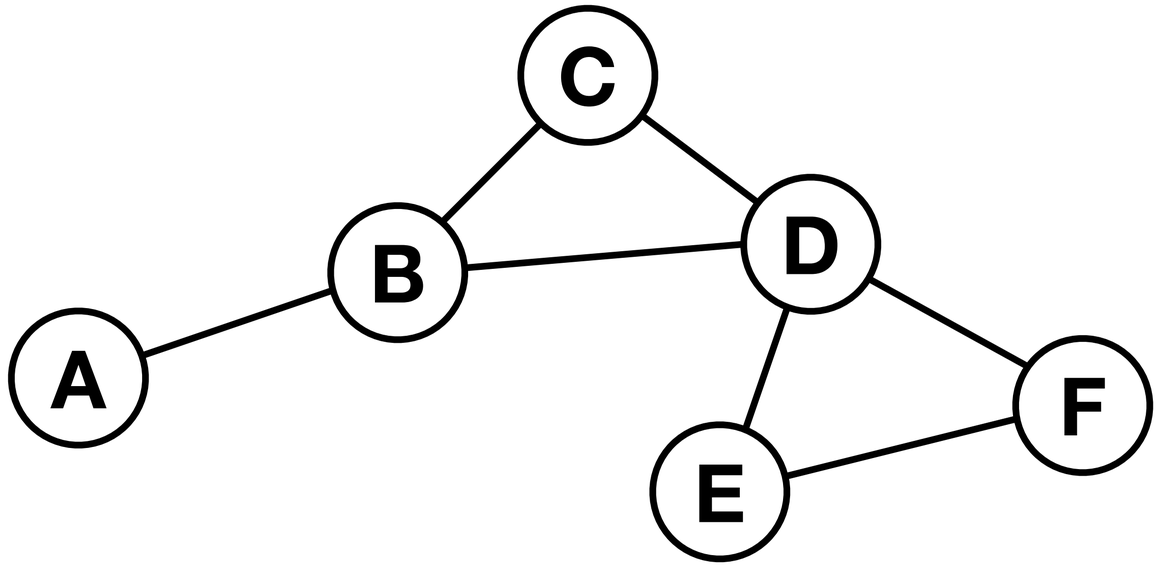
- Short introduction to graph deep learning
- Three strategies to make graphs multimodal
- Throughout: linking to previously introduced concepts of embeddings, encoders, and equivariance

GRAPH AS LIST OF EDGES

A graph $G = (V, E)$ is defined as a set of nodes $v_i \in V$ connected by edges $(v_i, v_j) \in E$

$$V = \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}\}$$

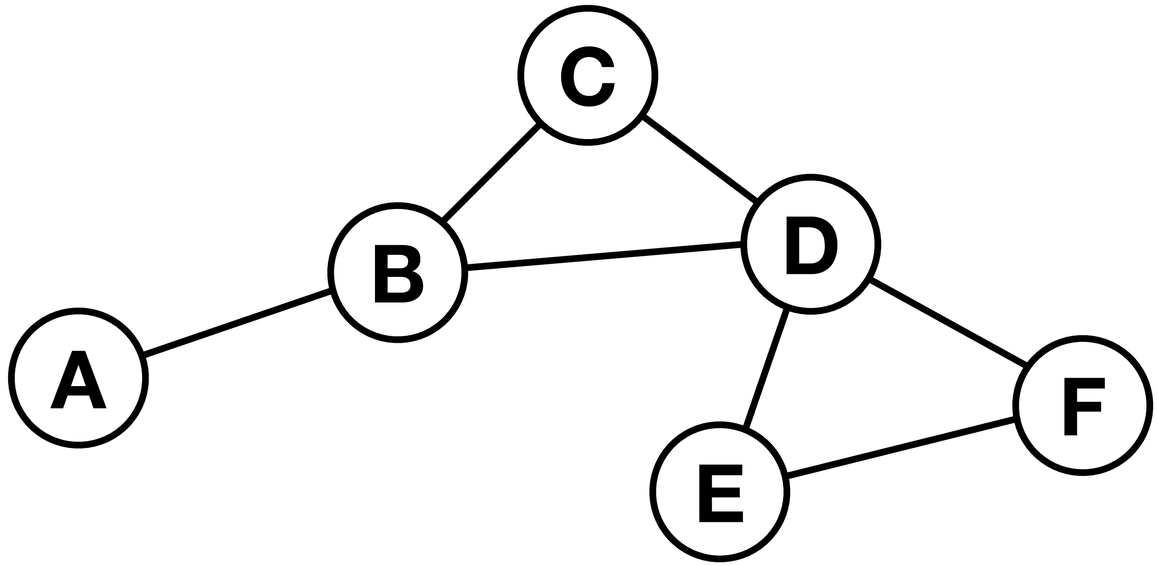
$$E = \{(\mathbf{A}, \mathbf{B}), (\mathbf{B}, \mathbf{C}), (\mathbf{B}, \mathbf{D}), (\mathbf{C}, \mathbf{D}), (\mathbf{D}, \mathbf{E}), (\mathbf{D}, \mathbf{F}), (\mathbf{E}, \mathbf{F})\}$$



GRAPH AS ADJACENCY MATRIX

Adjacency matrix A can be used as alternative representation

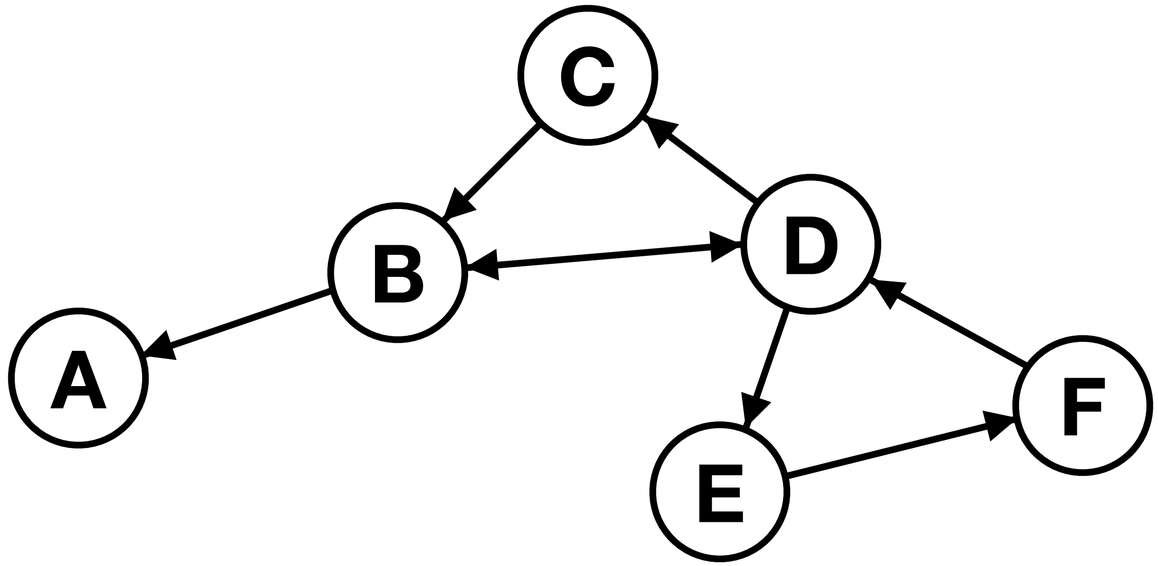
$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$



DIRECTED GRAPH

Edges can be undirected: $(v_i, v_j) = (v_j, v_i)$ or directed, $(v_i, v_j) \neq (v_j, v_i)$

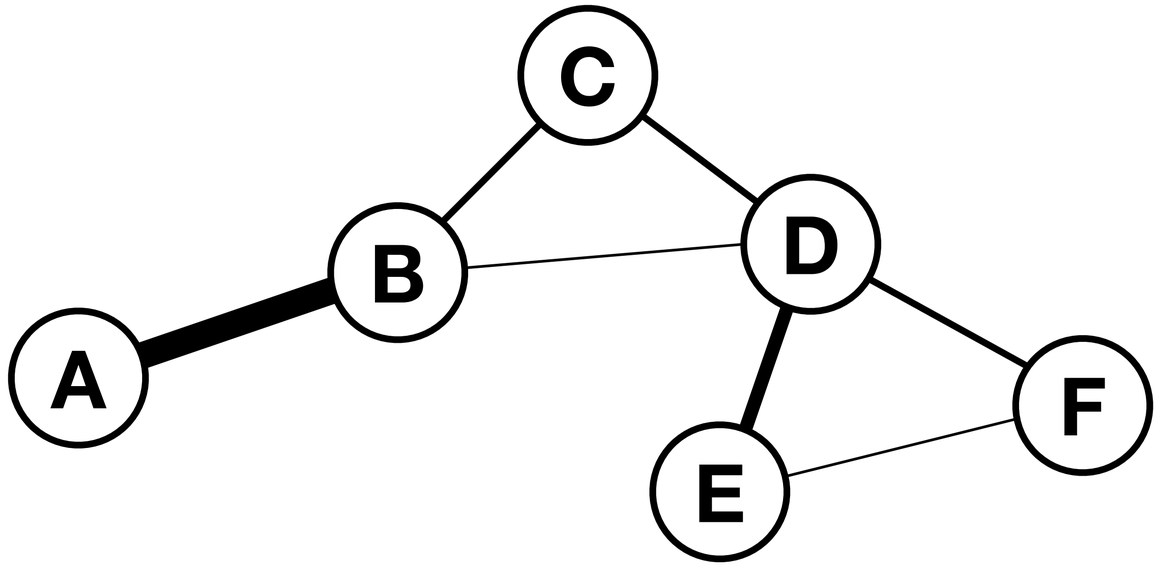
$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$



EDGE FEATURES

Edges can also be weighted or imbued with more complex embeddings

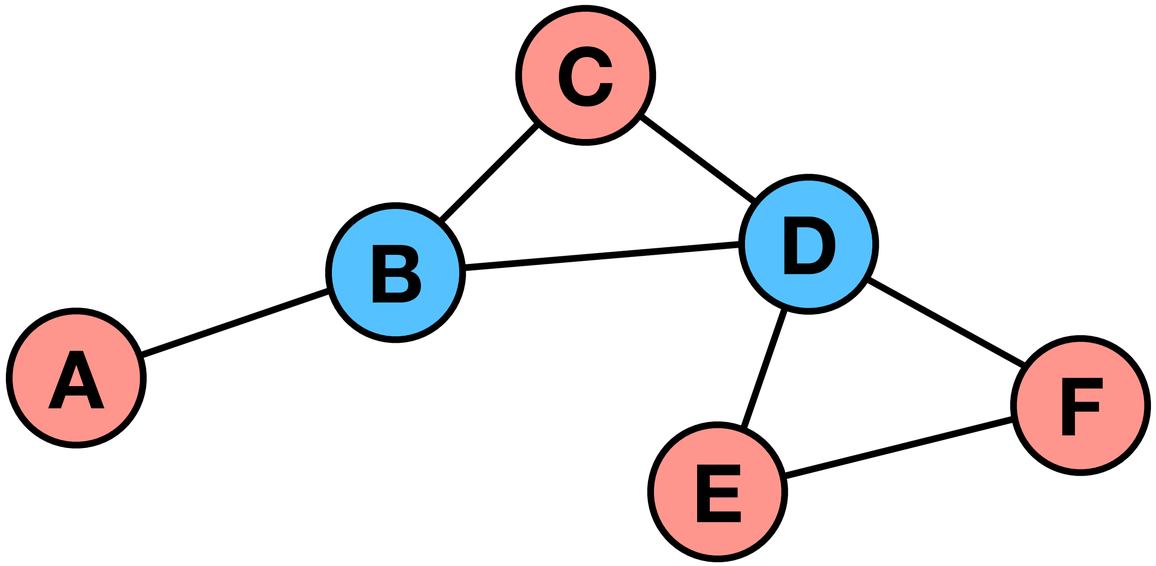
$$A = \begin{bmatrix} 0 & 2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0.5 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0.5 & 1 & 0 & 1.5 & 1 \\ 0 & 0 & 0 & 1.5 & 0 & 0.5 \\ 0 & 0 & 0 & 1 & 0.5 & 0 \end{bmatrix}$$



NODE FEATURES

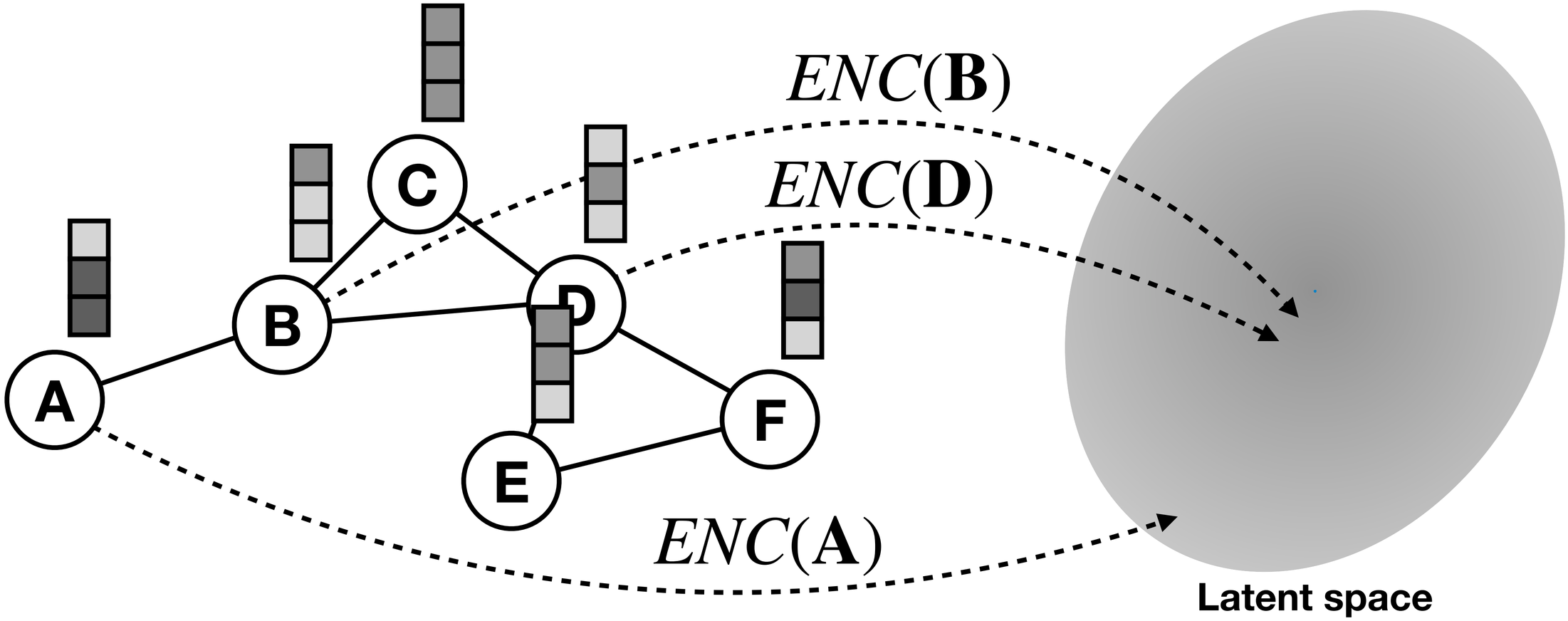
Each node is imbued with a d-dimensional feature vector

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \quad F = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$



NODE FEATURES

Mapping of nodes to latent space: $sim(\mathbf{v_i}, \mathbf{v_j}) \sim \mathbf{z_{v_i}}^T \mathbf{z_{v_j}}$

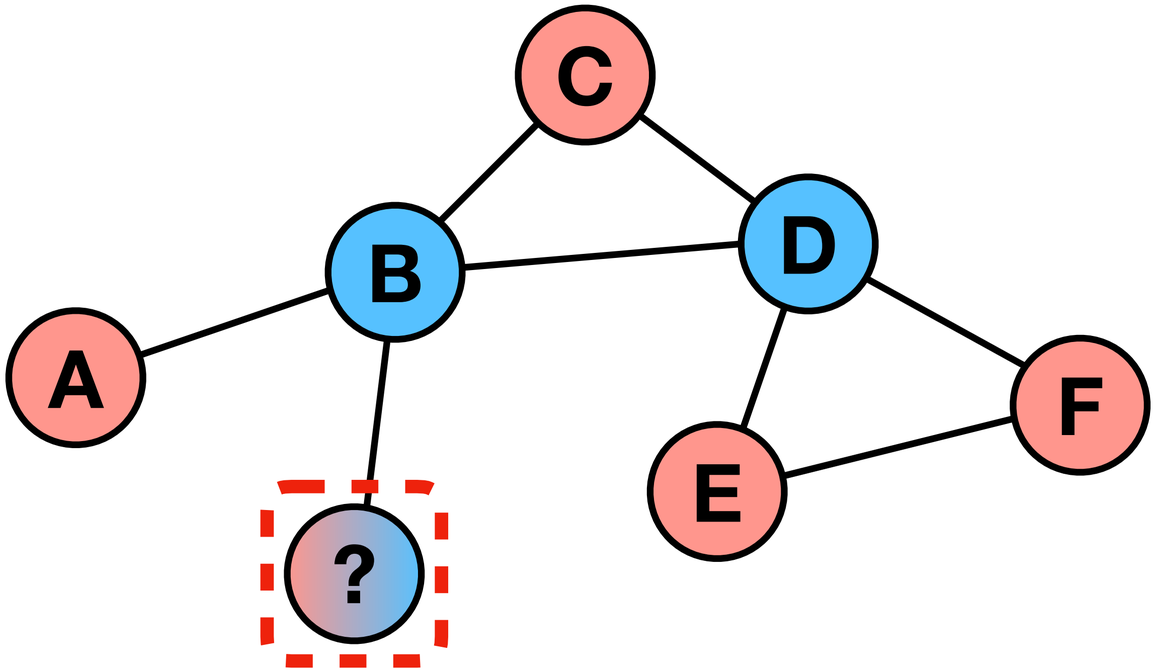


CLASSICAL NODE EMBEDDING STRATEGIES

- Feature engineering based on prior knowledge
  or graph structure (number of neighbors, node centrality, ...)
- Kernel-based methods with pre-defined graph kernels
- Spectral methods based on Laplacian eigenmaps
- Random walk based strategies (node2vec, ...)

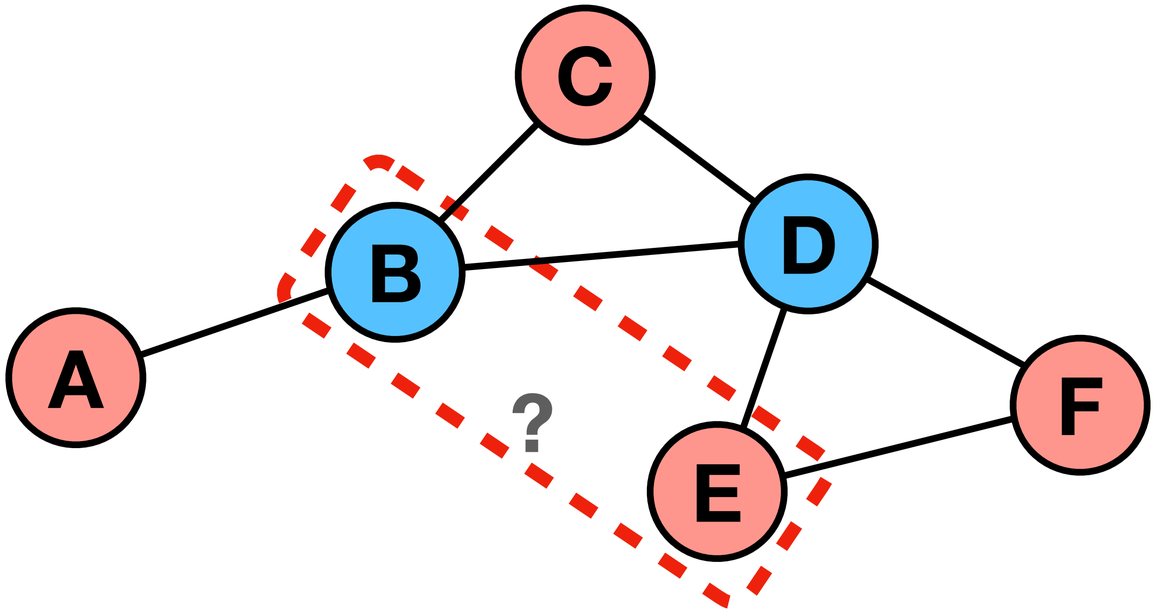
NODE-LEVEL PREDICTION

Evaluation of node features to predict node class



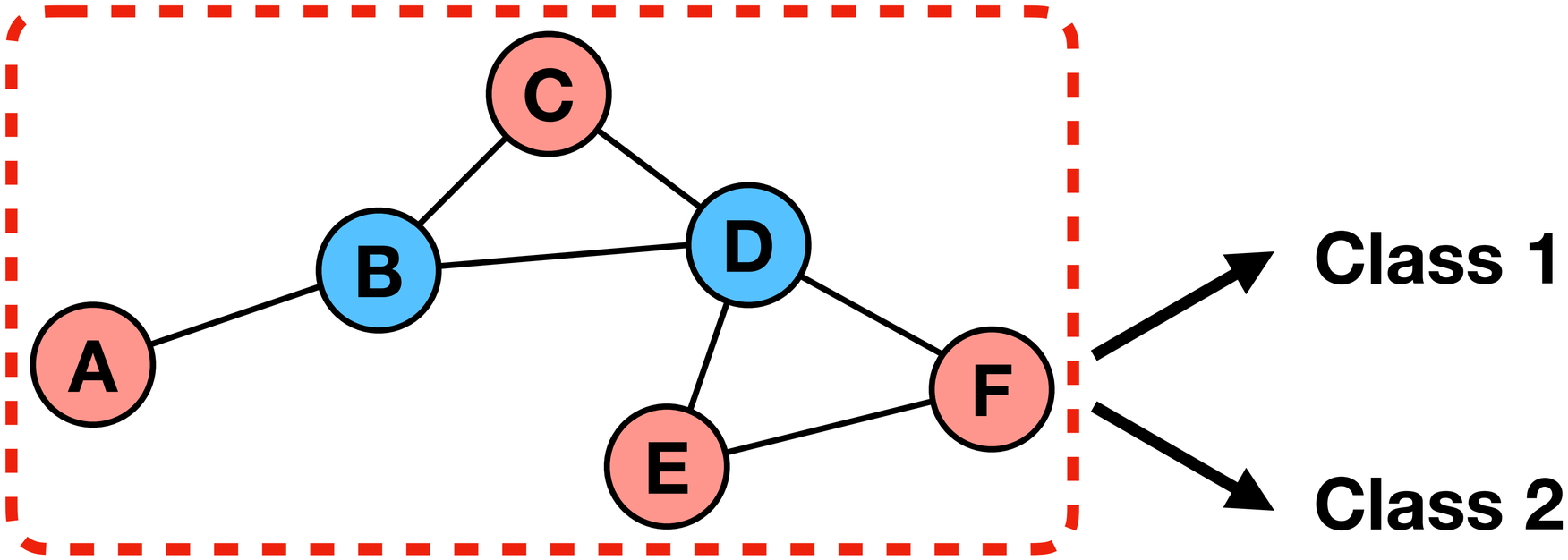
EDGE-LEVEL PREDICTION

Evaluation of edge features (e.g. features of neighboring nodes)
to predict existence or weight of potential edge



GRAPH-LEVEL PREDICTION

Evaluation of graph features (e.g. by aggregating all node features)
to make graph-level prediction



NODE FEATURES

- Generation of **node features by mapping of nodes to a latent space**
  - Similar nodes should obtain similar embeddings,
  - Dissimilar nodes should obtain distant embeddings
- Node features allow us to solve diverse downstream tasks
- Classical node embedding strategies may lack expressiveness

Can we use deep learning to derive useful node embeddings?

REMINDER: WEEK 2

LEARNING NON-LINEAR EMBEDDINGS

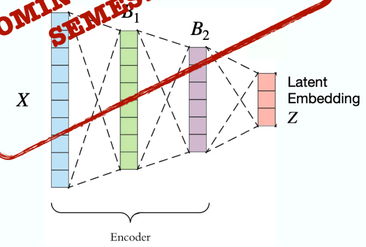
💡 To capture **even more** complex semantic relationships, learning **non-linear** embeddings is required.

Neural Network Encoders

Modern encoders learn embeddings through hierarchical, structured, and compositional non-linear transformations.

- **Note:** single-layer MLP without activation is simply a linear mapping ($B = AX + \beta$).
- Each intermediate hidden representation in an encoder is an embedding.

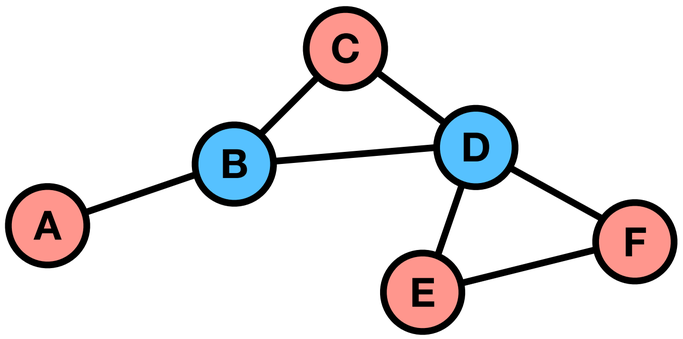
COMING LATER THIS SEMESTER !



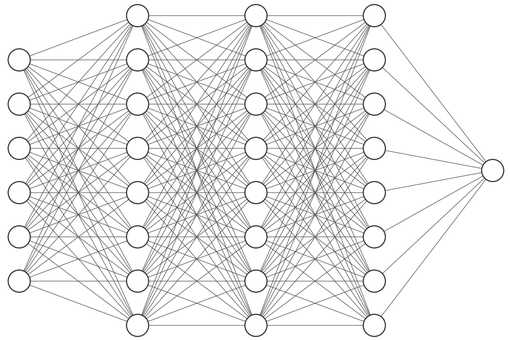
[7] Arden Dertat. https://medium.com/towards-data-science/applied-deep-learning-part-3-autoencoders-1c083af4d798 *(accessed January 2025)*.

A NAIVE APPROACH

Represent graph as adjacency and feature matrix
and process with a multi-layer perceptron



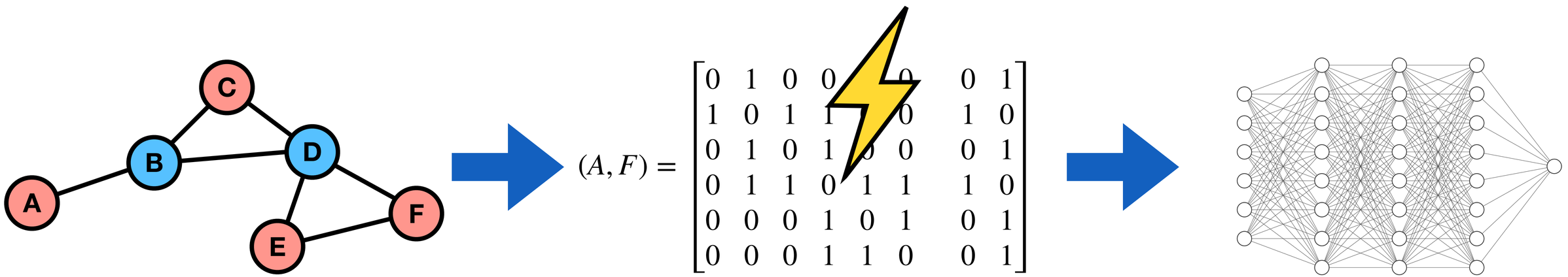
$$(A, F) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$



A NAIVE APPROACH

Problem #1:

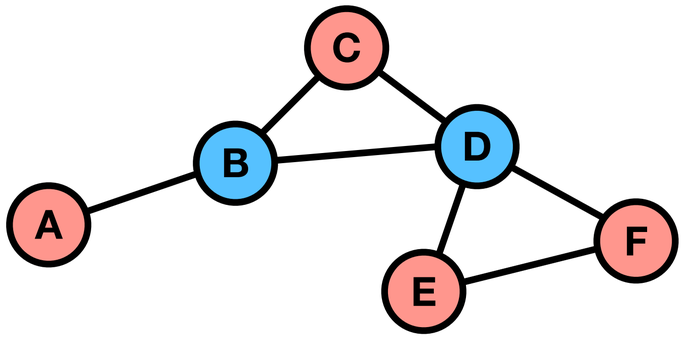
- Adjacency matrices can be large and sparse
- Number of parameters scales with $O(|V|)$



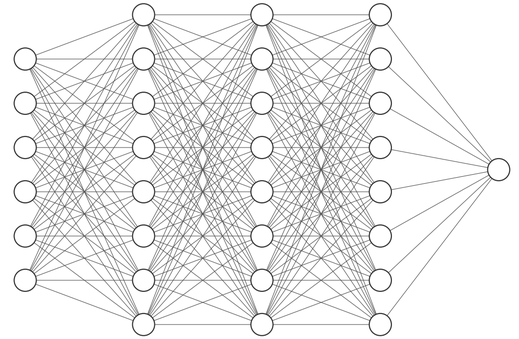
A NAIVE APPROACH

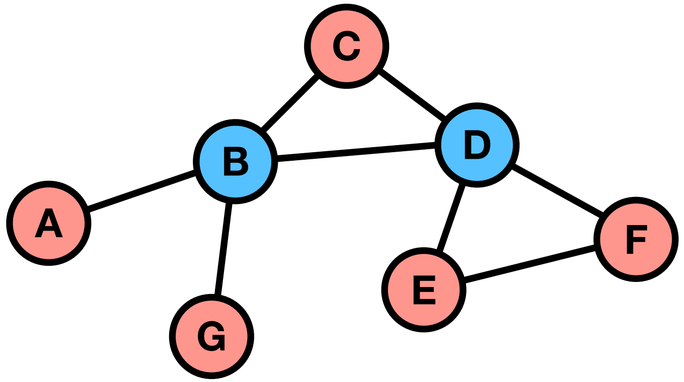
Problem #2:

- Graphs may have different sizes
- MLPs cannot process unseen graph configurations ("transductive)

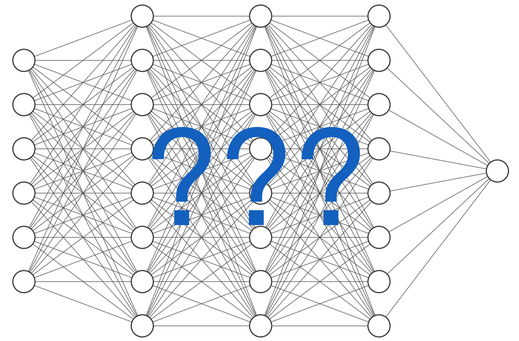


$$(A, F) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$





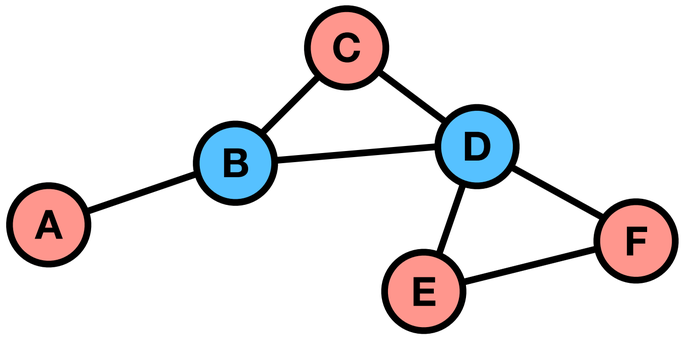
$$(A, F) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$



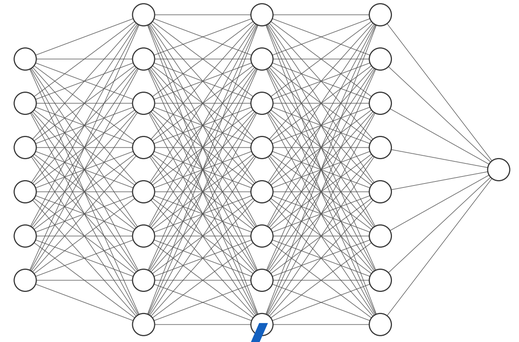
A NAIVE APPROACH

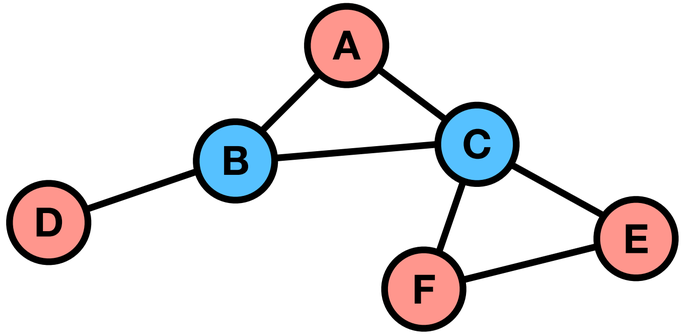
Problem #3:

- There exist $|V|!$ permutations of graph orders but no canonical ordering
- MLPs are not permutation equivarant

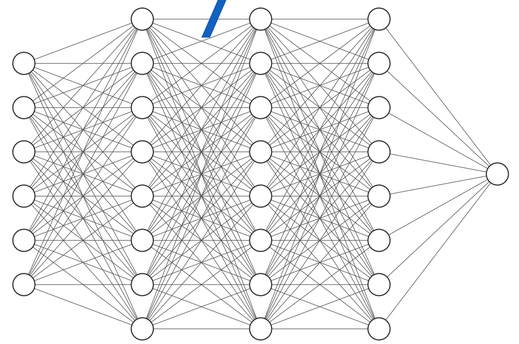


$$(A, F) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$





$$(A, F) = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$



A NAIVE APPROACH

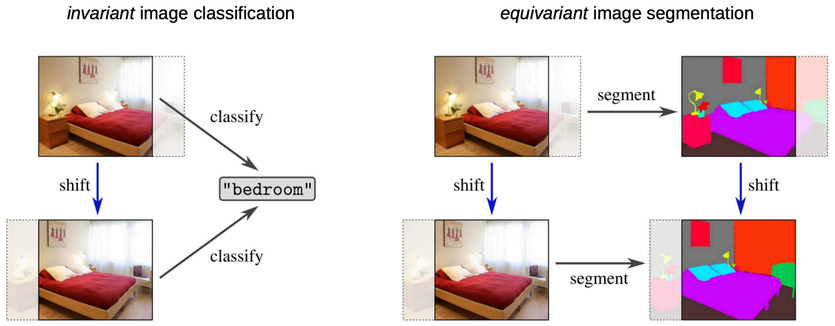
- Problems
  - Number of parameters scales with $O(|V|)$
  - MLPs cannot process unseen graph configurations
  - MLPs are not permutation equivarant

We need to design a neural network that is permutation equivariant

REMINDER: WEEK 4

INVARIANCE VS. EQUIVARIANCE

🤔 How do we include this information into our model architectures?

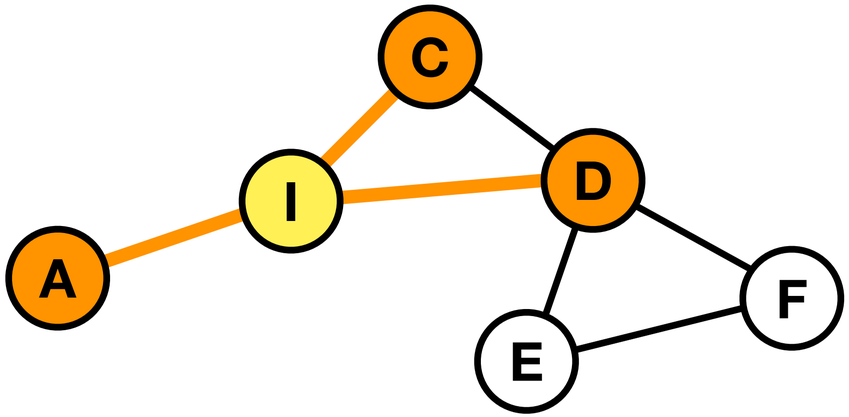


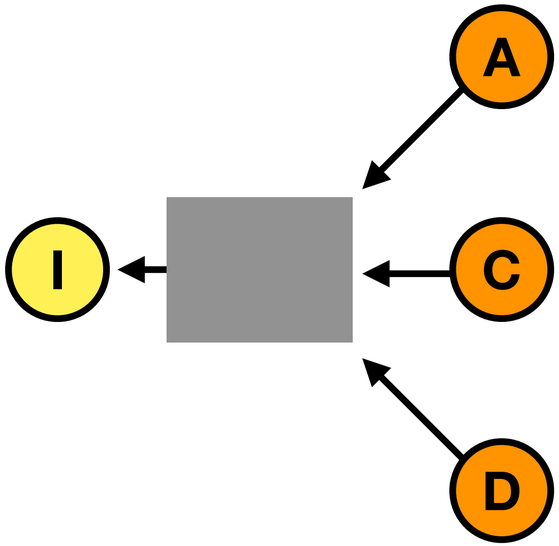
Intuitively, invariance loses information while equivariance preserves it.

[1] https://www.sci.unich.it/geodeep2022/slides/Groups_Representations_and_Equivariance.pdf

GRAPH CONVOLUTIONAL NEURAL NETWORKS

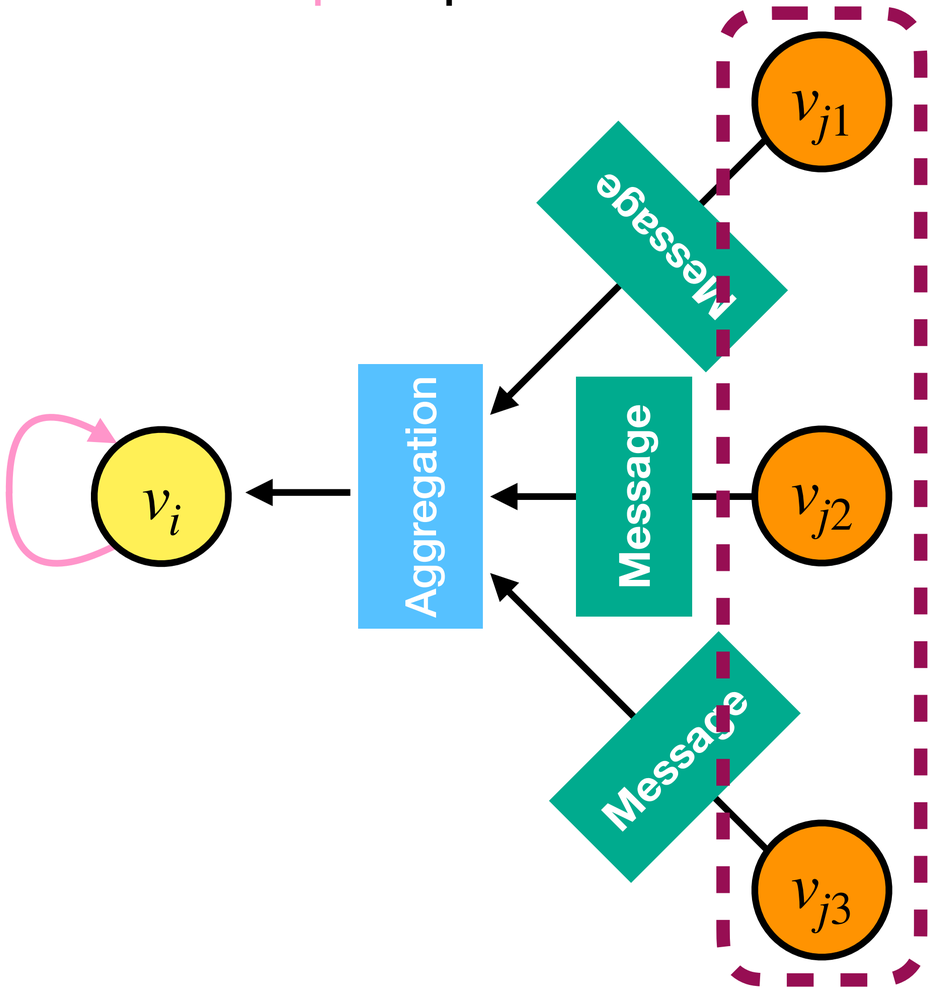
- Step 1: determine node computation graph based on graph structures
- Step 2: propagate and transform information





GRAPH CONVOLUTIONAL NEURAL NETWORKS

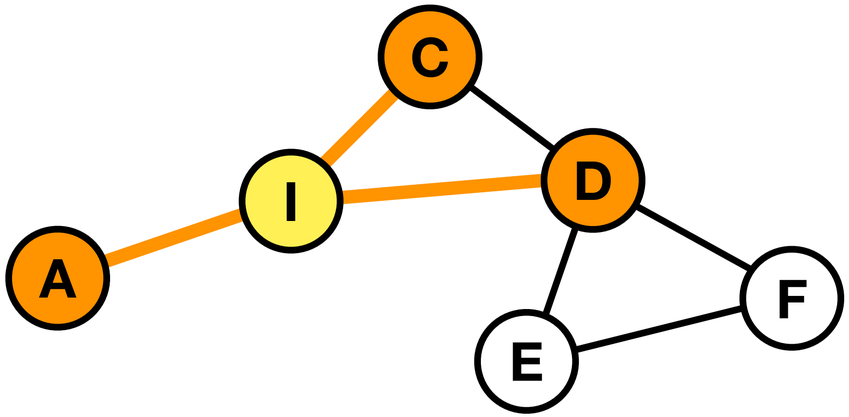
- Message passing and subsequent aggregation of neighboring nodes
- Self-loop to preserve content of updated node

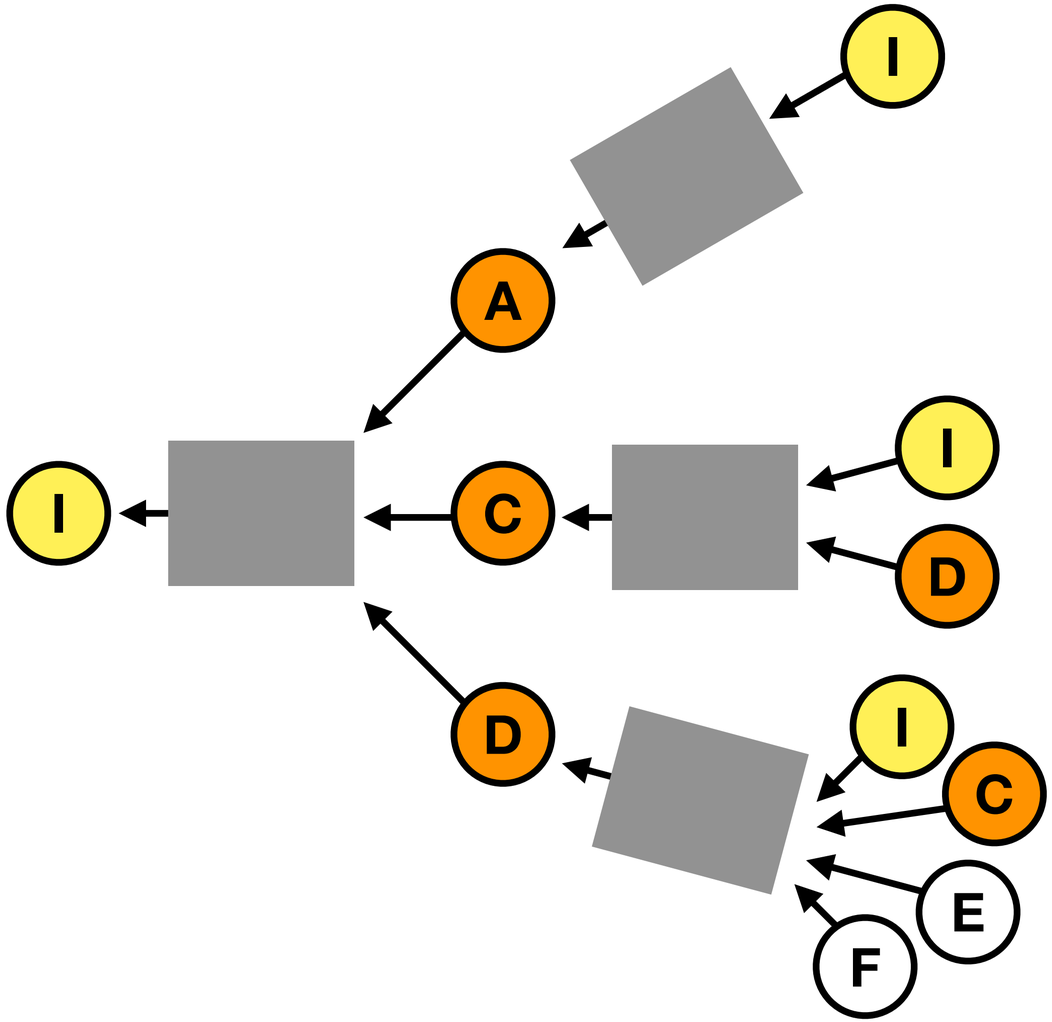


$$v_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}_i} \frac{1}{c_i} W^{(l)} v_j^{(l)} + W_0^{(l)} v_i^{(l)}\right)$$

GRAPH CONVOLUTIONAL NEURAL NETWORKS

- Multiple layers can be stacked





AGGREGATION FUNCTION

- With increasing number of graph convolutional layers the receptive field grows exponentially
- Typical pooling functions:
  - $Mean(\cdot)$
  - $Sum(\cdot)$
  - $Max(\cdot)$
- Advanced network architectures use attention mechanisms

REMINDER: WEEK 4

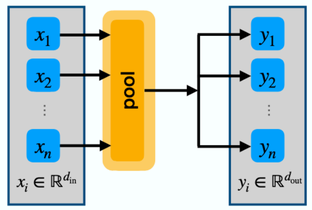
POOL AND DISTRIBUTE

Pool and Distribute

1. Pool inputs to single vector (e.g. using mean/max pooling)
2. Copy pooled vector for each element

Optional

- Apply MLP after pool / before distribute
- Add outputs to
  - residual connections of x
  - or element-wise operations on x



	y_1	y_2	...	y_n
x_1	1	1	1	1
x_2	1	1	1	1
...	1	1	1	1
x_n	1	1	1	1

Information Flow

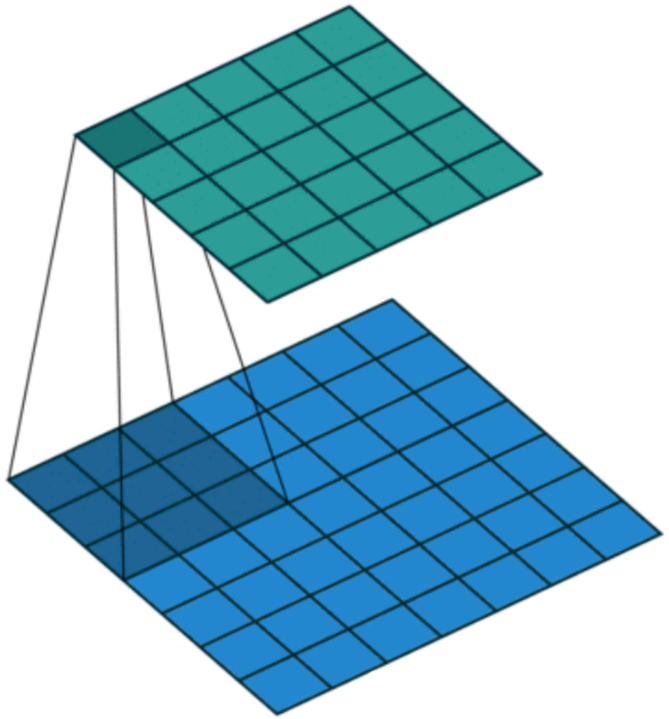
GNN SUBSUME TRANSFORMER

Transformers are graph neural networks - To be shown in the homework

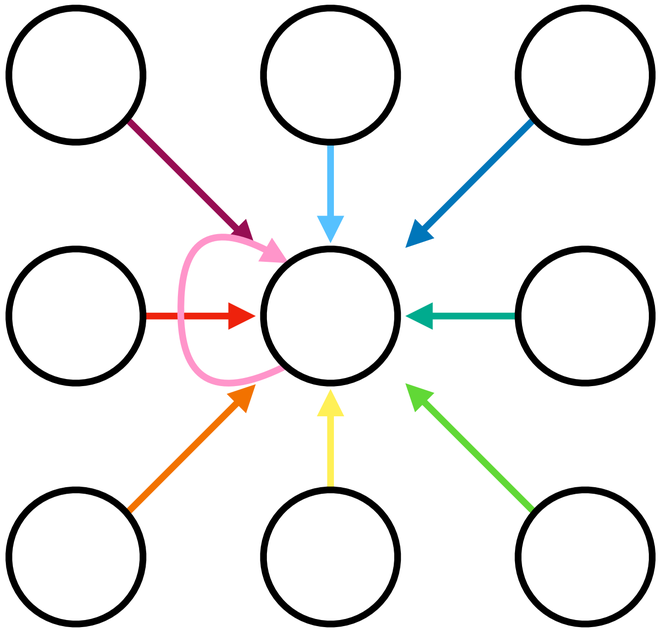
GNNS SUBSUME CNNS

Images and sequences can be seen as special cases of graphs

CNNs can be seen as a special GNN with fixed neighborhood size and ordering



3x3 convolution

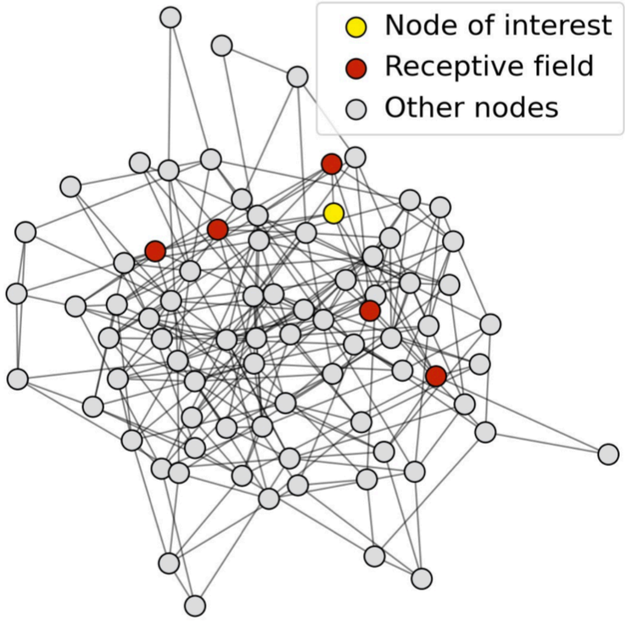


3x3 convolution as graph

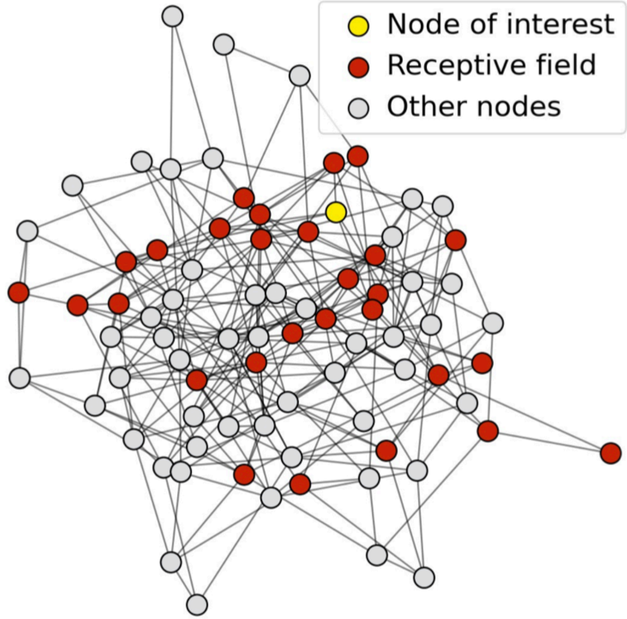
GRAPH OVERSMOOTHING

With increasing number of graph convolutional layers
the receptive field grows exponentially

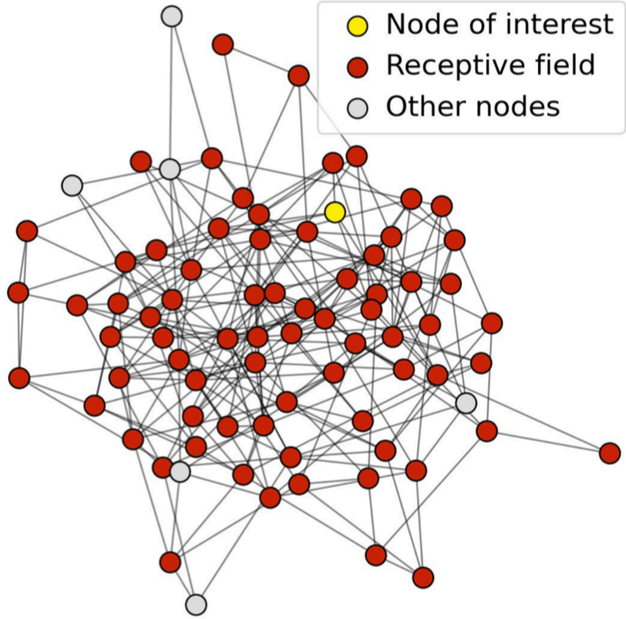
Receptive field for
1-layer GNN



Receptive field for
2-layer GNN



Receptive field for
3-layer GNN

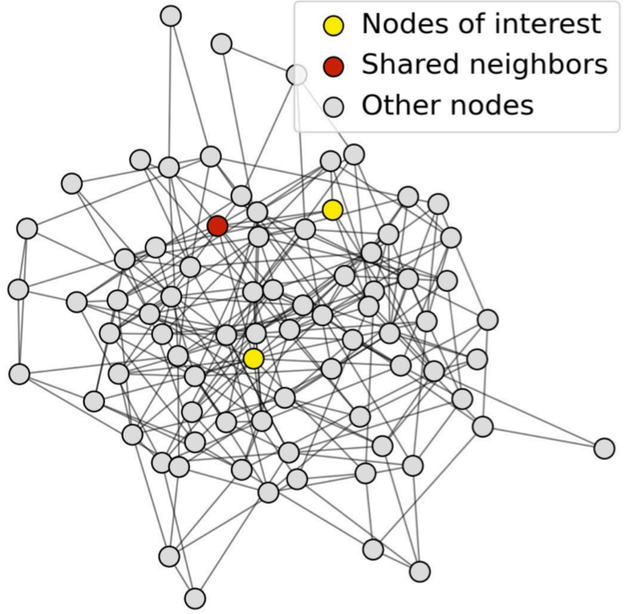


GRAPH OVERSMOOTHING

As the receptive fields of all nodes overlap, they become highly similar, leading to degenerate node embeddings

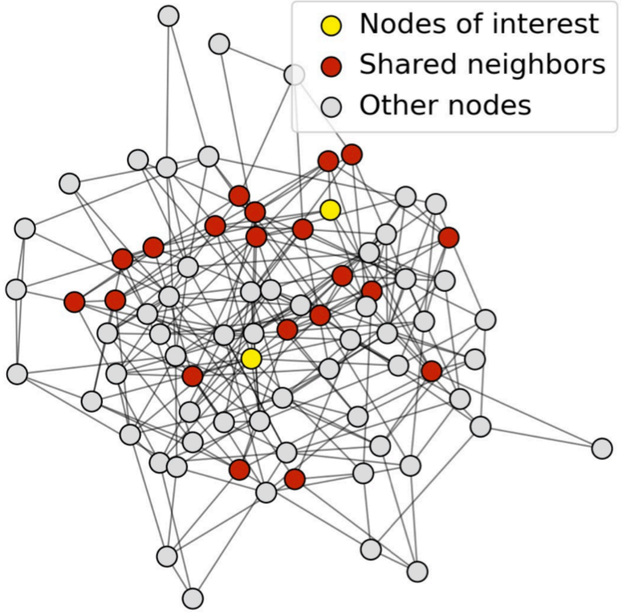
1-hop neighbor overlap

Only 1 node



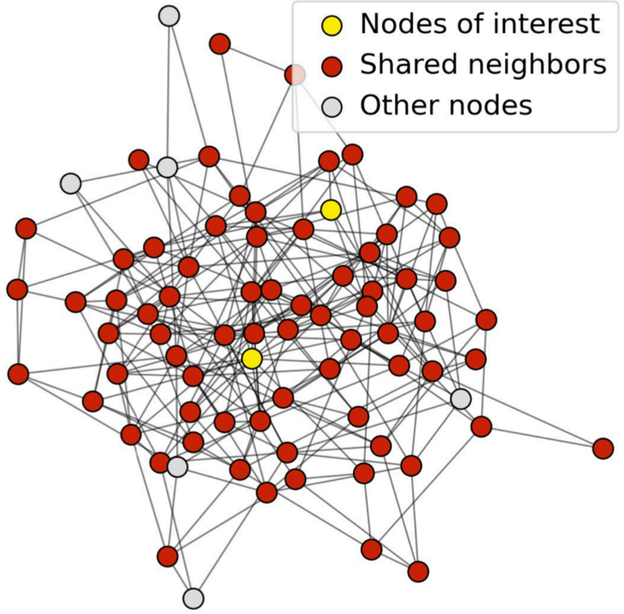
2-hop neighbor overlap

About 20 nodes



3-hop neighbor overlap

Almost all the nodes!



SUMMARY

Strengths of graph deep learning:

- Explicit inductive bias
- Light-weight
- Parameter-efficient
- Flexible with missing modalities

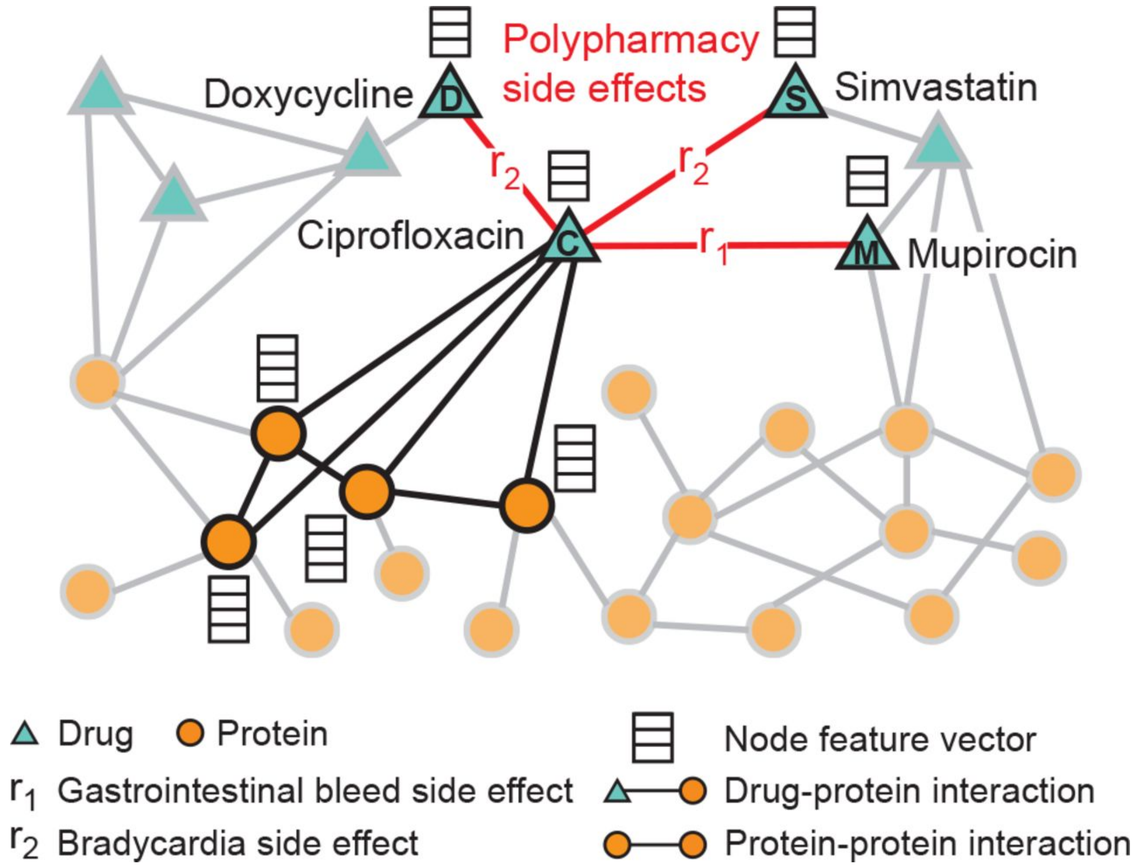
Weaknesses:

- Dependent on graph quality
- Limited long-range dependency modeling
- Cumbersome implementation
- Lack of scalability

GRAPHS IN MULTIMODAL DEEP LEARNING

- Graphs are commonly found in **multimodal deep learning**
  - Graph as **input** of a multimodal pipeline
  - Graph as **output** of a multimodal pipeline
  - Graph as **structure to relate** multimodal data embeddings

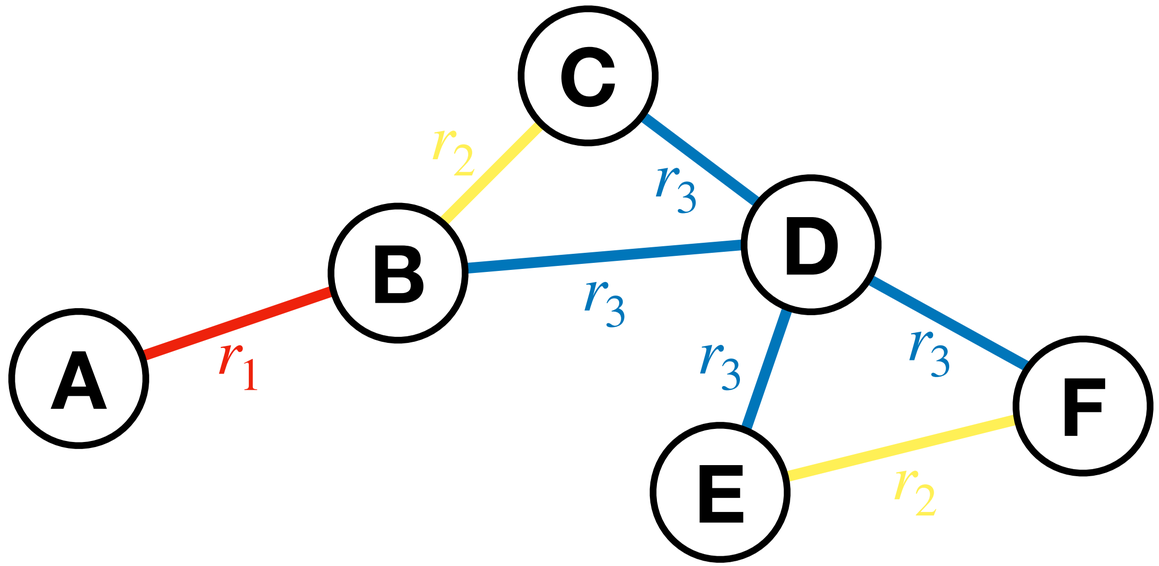
HETEROGENEOUS GRAPHS



Drug interaction graph

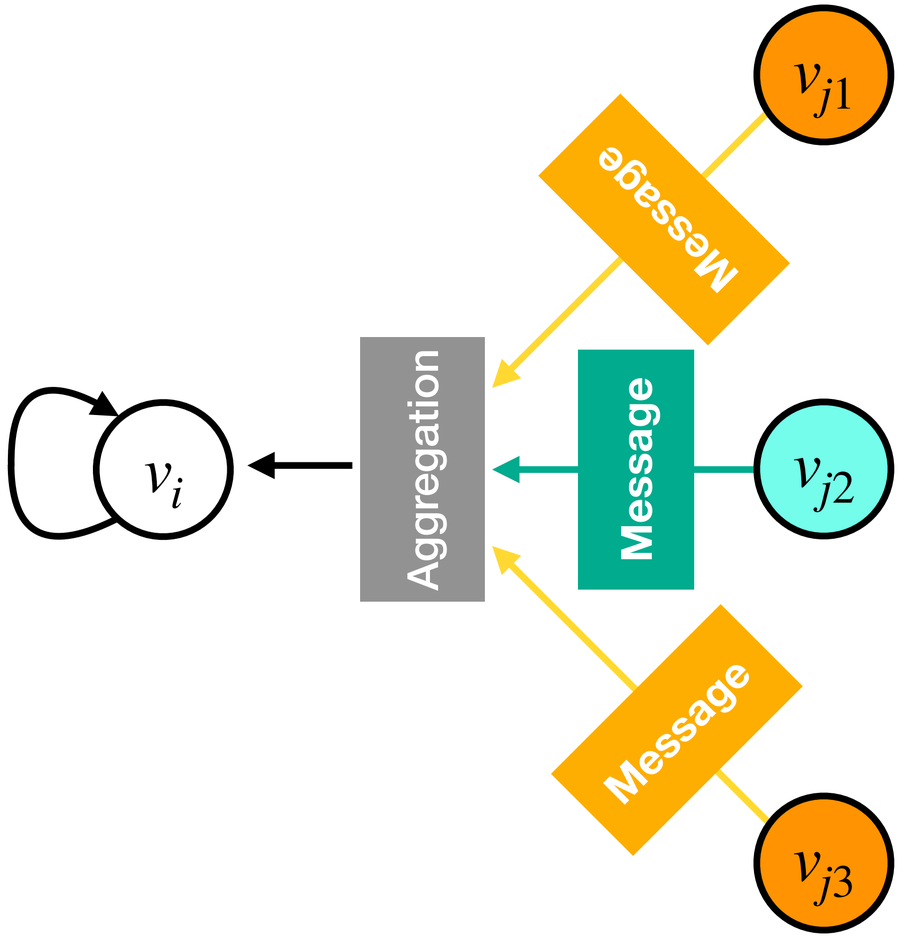
HETEROGENEOUS GRAPHS

Each nodes have a type $\tau(v)$, each edge has a type $\phi(v_i, v_j)$,
resulting in a relation type for each edge $r(v_i, v_j) = (\tau(v_i), \phi(v_i, v_j), \tau(v_j))$



RELATIONAL GRAPH CONVOLUTIONAL NEURAL NETWORKS

Use of different message weight matrices depending on the type of relation



$$v_i^{(l+1)} = \sigma\left(\sum_{r\in R}\sum_{j\in\mathcal{N}_i^r}\frac{1}{c_{i,r}}W_r^{(l)}v_j^{(l)} + W_0^{(l)}v_i^{(l)}\right)$$

RELATIONAL GRAPH CONVOLUTIONAL NEURAL NETWORKS

- Rapid growth of parameters with number of relations
  - Overfitting becomes an issue
- Potential mitigation strategies
  - Use of sparse (block diagonal) weight matrices
  - Sharing of weights across different message matrices

FURTHER MATERIAL

- TUM lecture Machine Learning for Graphs and Sequential Data (IN2323)
- Stanford Machine Learning with Graphs lecture by Jure Leskovec (https://web.stanford.edu/class/cs224w)
- Graph Representation Learning Book by William L. Hamilton (https://www.cs.mcgill.ca/~wlh/grl_book)